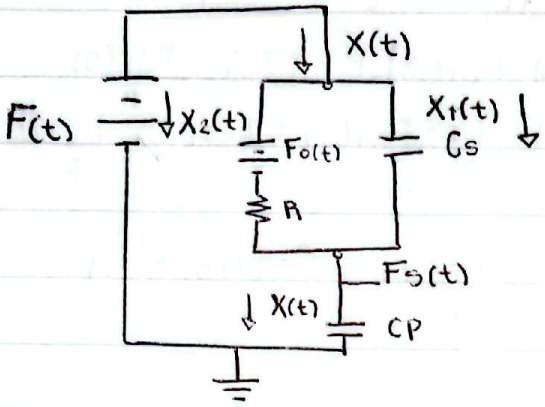
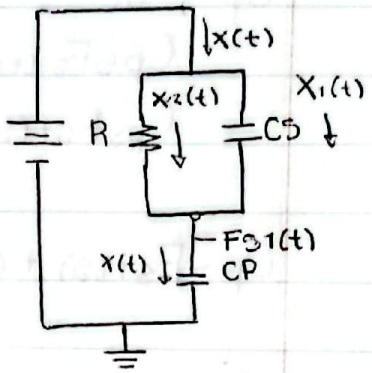
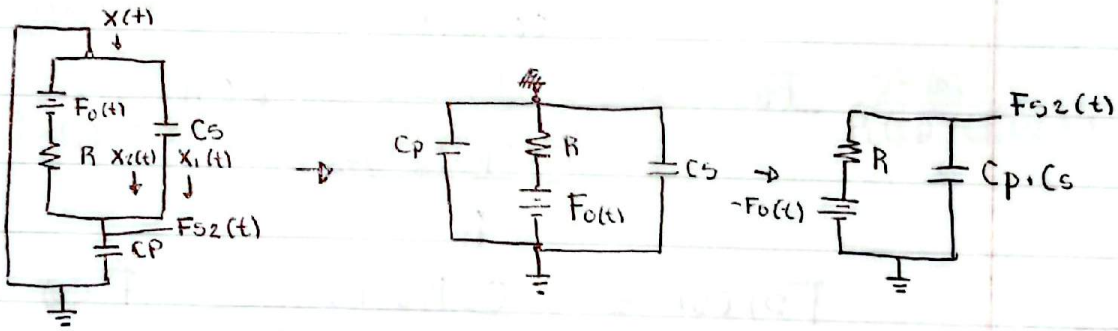
$X(t) = X_1(t) + X_2(t)$

Variante 1 sin $F_o(t)$





Variante 2 sin $F(t)$



Analisis matematico.

Debido a que el sistema contiene dos fuentes de voltaje se debe aplicar el principio de superposición

Circuito apagando $F_o(t)$

$$X(t) = X_2(t) + X_2(t)$$

$$X_2(t) = C_p \frac{dF_{s1}(t)}{dt}$$

$$X_{s1}(t) = C_s \frac{d[F(t) - F_{s1}(t)]}{dt}$$

Escaneado con CamScanner

$$x_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p s F_{s1}(t) = C_s s [F(s) - F_{s1}(s)] + \frac{F(s) - F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) = C_s s F(s) - C_s s F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) + C_s s F_{s1}(s) + \frac{F_{s1}(s)}{R} = C_s s F(s) + \frac{F(s)}{R}$$

~~$C_p s F_1(s) +$~~

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{s1}(s) = \frac{\frac{C_s R s + 1}{R}}{\frac{R(C_p + C_s)s + 1}{R}} F(s)$$

$$F_{s1}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$$

Apagando $F(t)$

$$-aF(t) = Rx(t) + \frac{1}{C_p + C_s} \int x(t)dt$$

$$F_{s2}(t) = \frac{1}{C_p + C_s} \int x(t)dt$$

$$-aF(s) = Rx(s) + \frac{x(s)}{(C_p + C_s)s}$$

$$F(s) = -\frac{1}{a}\left(Rx(s) + \frac{x(s)}{(C_p + C_s)s}\right)$$

Escaneado con CamScanner

$$\frac{F_{s2}(s)}{F(t)} = \frac{\frac{X(s)}{(C_p+C_s)s}}{-\frac{1}{a}\,\frac{R(C_p+C_s)s+1}{(C_p+C_s)s}\,X(s)}$$

$$= \frac{-\frac{1}{a}\,\frac{R(C_p+C_s)s+1}{(C_p+C_s)s}\,X(s)}{}$$

$$\frac{F_{s2}(s)}{F(t)} = \frac{\frac{X(s)}{(C_p+C_s)s}}{-\frac{1}{a}\,\frac{R(C_p+C_s)s+1}{(C_p+C_s)s}\,X(s)} = \frac{a}{R(C_p+C_s)s+1}$$

$$F_{s2}(s) = -\frac{aF(s)}{R(C_p+C_s)s+1}$$

Función de transferencia

$$F_s(s) = F_{s1}(s) + F_{s2}(s) = \frac{C_sR_s+1}{R(C_p+C_s)s+1}F(s) - \frac{aF(s)}{R(C_p+C_s)s+1}$$

$$F_s(s) = \frac{C_sR_s+1-a}{R(C_p+C_s)s+1}\,F(s)$$

$$F_s = F_s^1(s) + F_s^2(s) = \frac{(RC_ss+1)F(s)}{(RC_p+RC_s)s+1} - \frac{aF(s)}{(RC_p+RC_s)s+1}$$

$$F = \frac{RC_ss+1-a}{(RC_p+RC_s)s+1}\,F(t)$$

$$\frac{F_s(s)}{F(s)} = \frac{RC_ss+1-a}{(RC_p+RC_s)s+1}$$

Escaneado con CamScanner

$$e(s) = \lim_{s \to 0} sF(s)\left(1 - \frac{F_s(s)}{F(s)}\right) = \lim_{s \to 0} \frac{1}{s}\left(1 - \frac{RC_s s + 1 - a}{(RC_p + RC_s)s + 1}\right) = a$$

Lazo abierto

$$(RC_p + RC_s)s + 1 = 0$$

$$\lambda = -1/RCP + RCs$$

Escaneado con CamScanner